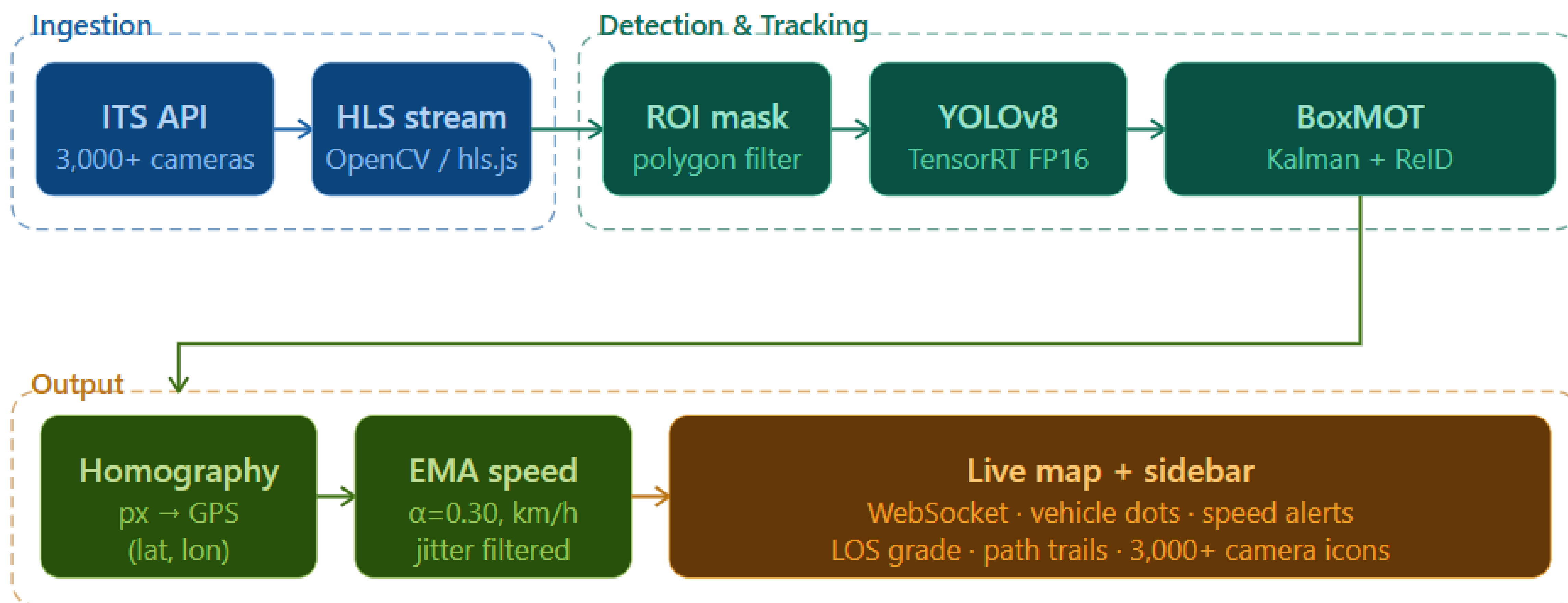




This pipeline ingests HLS camera streams, detects and tracks vehicles using YOLOv8 and BoxMOT, converts pixel positions to GPS coordinates, estimates smoothed speeds, and visualizes real-time traffic status on a live map.

INTRODUCTION

Korea's 3,000+ highway CCTV cameras generate 24/7 footage — yet real-time traffic state remains inaccessible. Traffic Digital Twin delivers GPS-accurate vehicle tracking with no additional hardware.

3,000+
live cameras

0
Extra Hardware

4pt
GPS calibration

- 📷 YOLOv8 + BoxMOT tracking — car, motorcycle, bus, truck
- 📖 Pixel → GPS homography (~130m × 70m FOV per camera)
- ⚙️ EMA speed ($\alpha=0.30$) · LOS grading A–F · bottleneck detection
- ✏️ Browser ROI editor and camera calibration — no server restart

TECH STACK

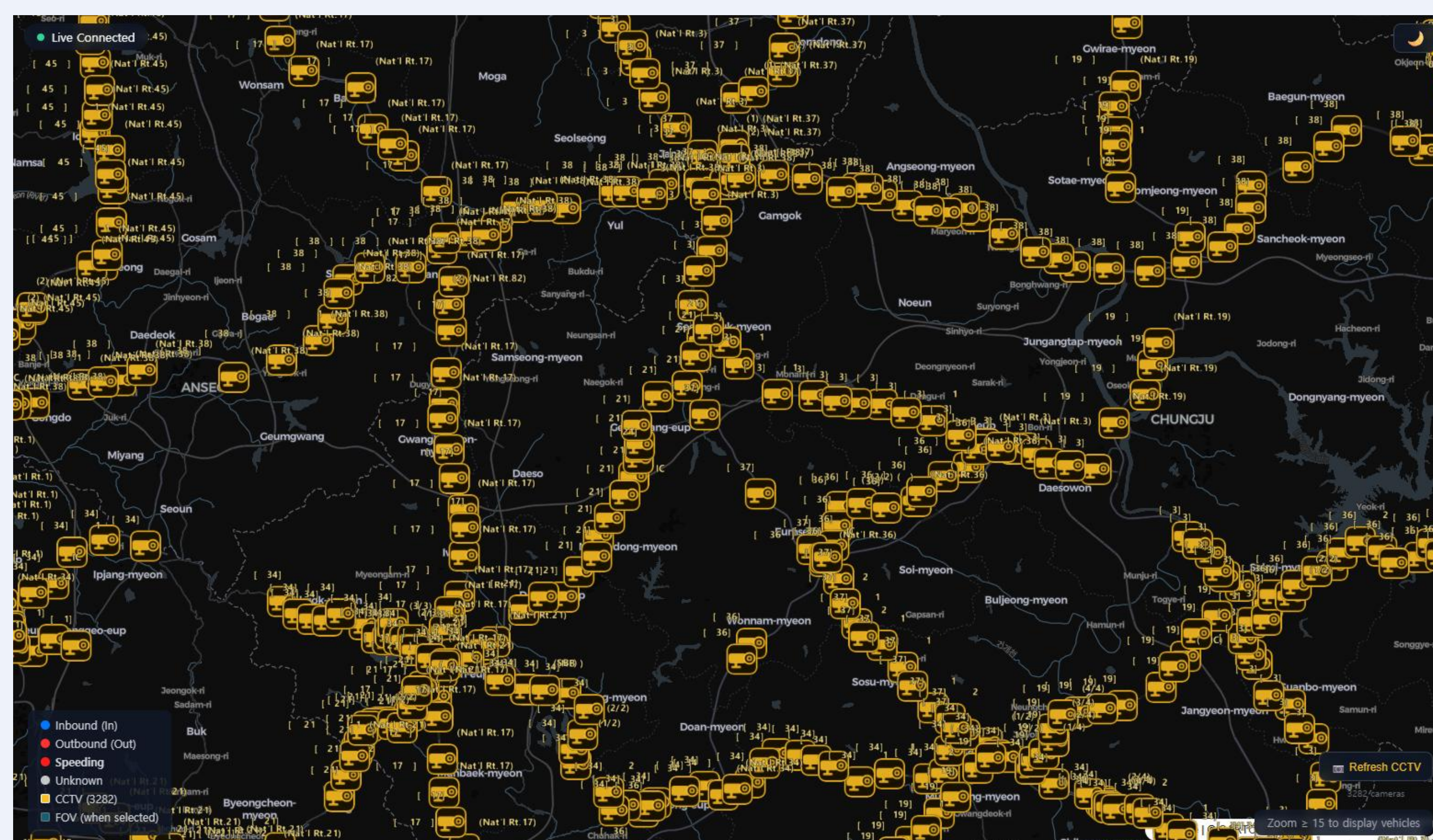
Backend

WebSocket
FastAPI
Python 3.11
TensorRT FP16
YOLOv8
OpenCV
BoxMOT
PyTorch 2.6

Frontend

React 18
hls.js
deck.gl
Recharts
MapLibre

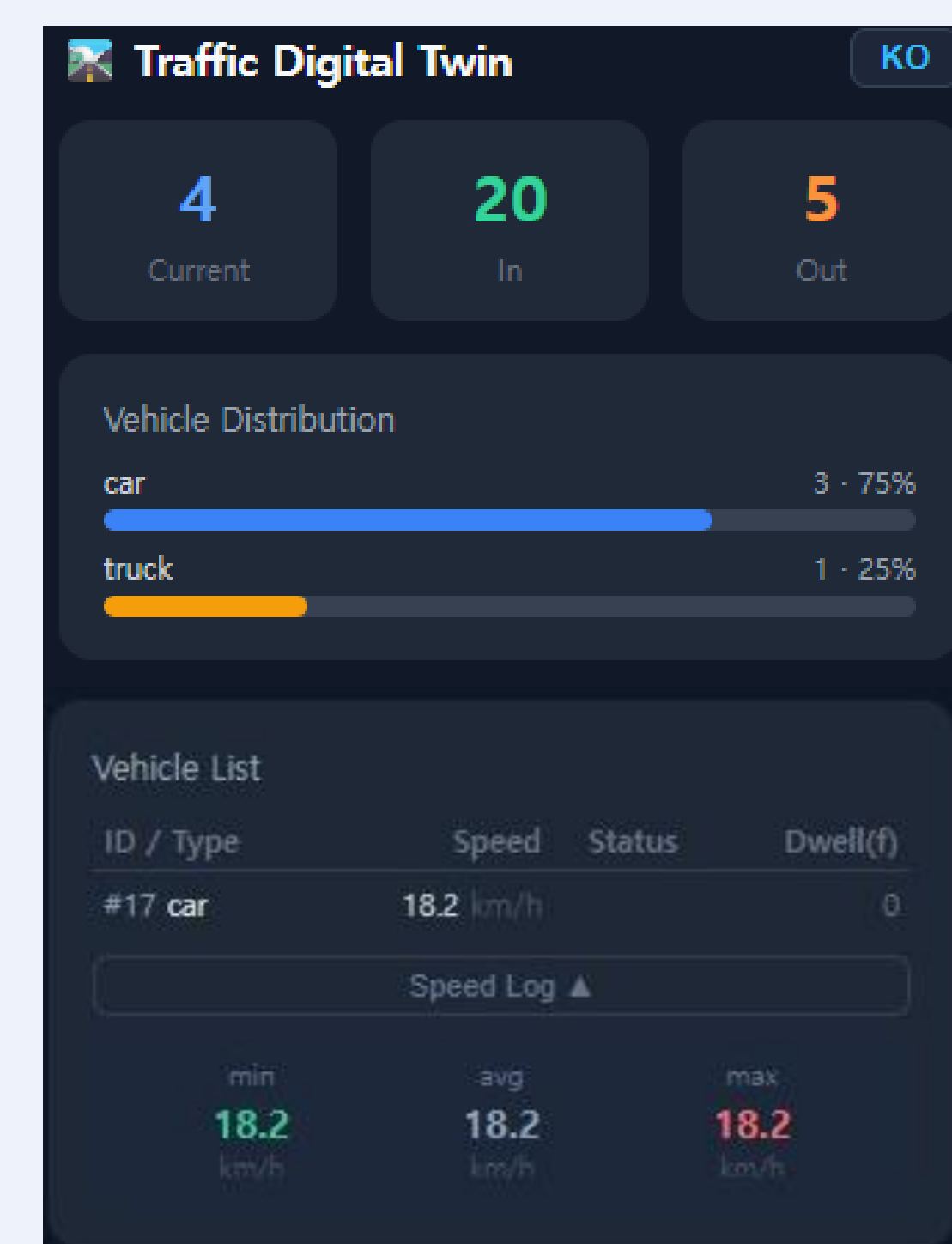
DEMO



① CCTV Map View



② YOLO Detection



③ Live Sidebar
— counts, vehicle list, speed log, LOS, alerts

RESULTS & LIMITAION

37fps

~27ms pipeline
YOLOv8x FP16

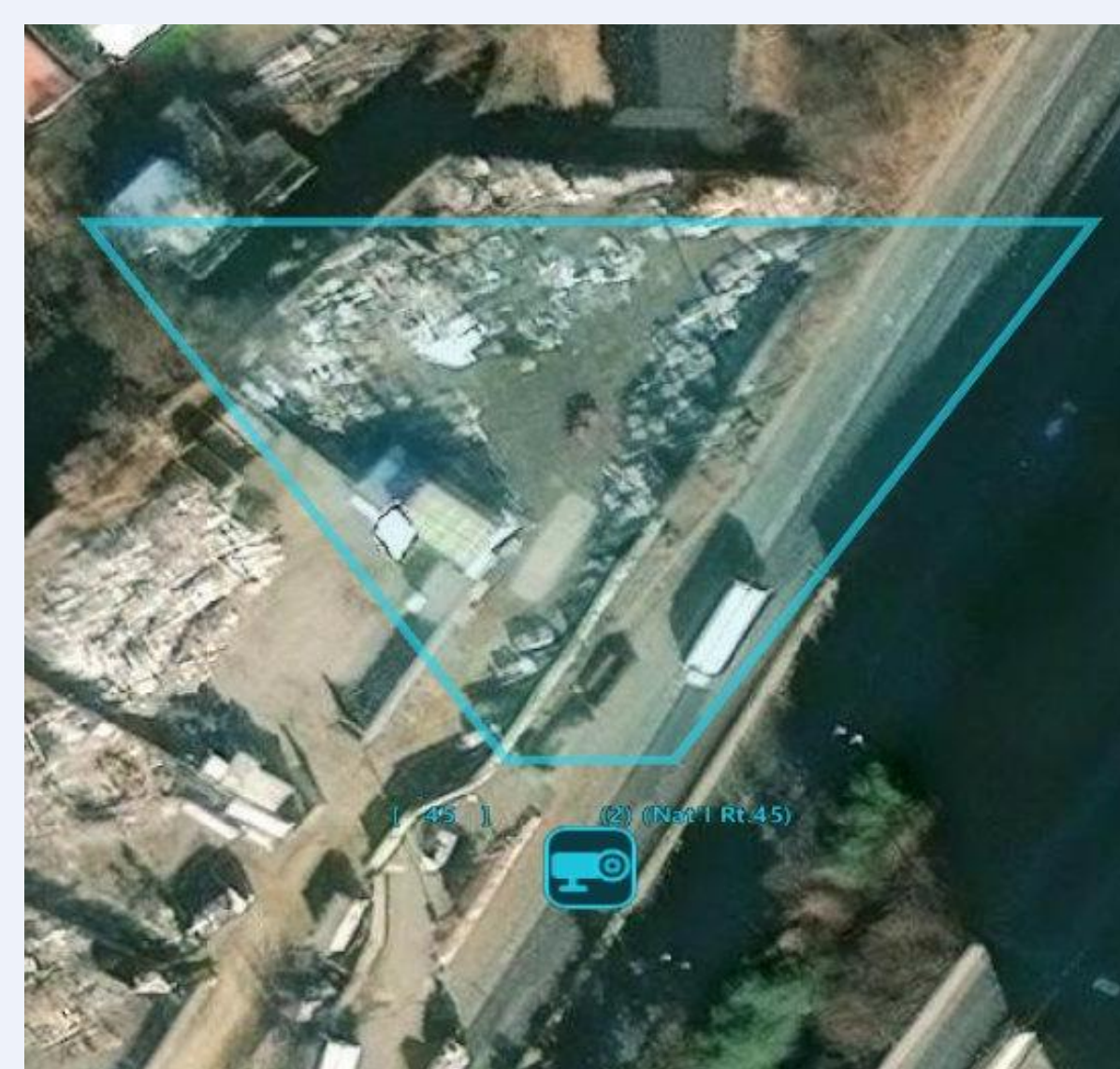
50fps

~20ms pipeline
YOLOv8s FP16

30–50% faster with FP16 vs FP32
<1% accuracy loss · both exceed 30fps threshold

Known Limitation

- ⚠️ Homography valid only inside calibrated quad
- ⚠️ Manual ROI required per camera
- ⚠️ Reduced accuracy in night / low-light scenes
- ⚠️ Speed threshold fixed at 120 km/h in config



④ Calibration
— satellite FOV



⑤ ROI Editor — green polygon